Cardiac Deformation Lab

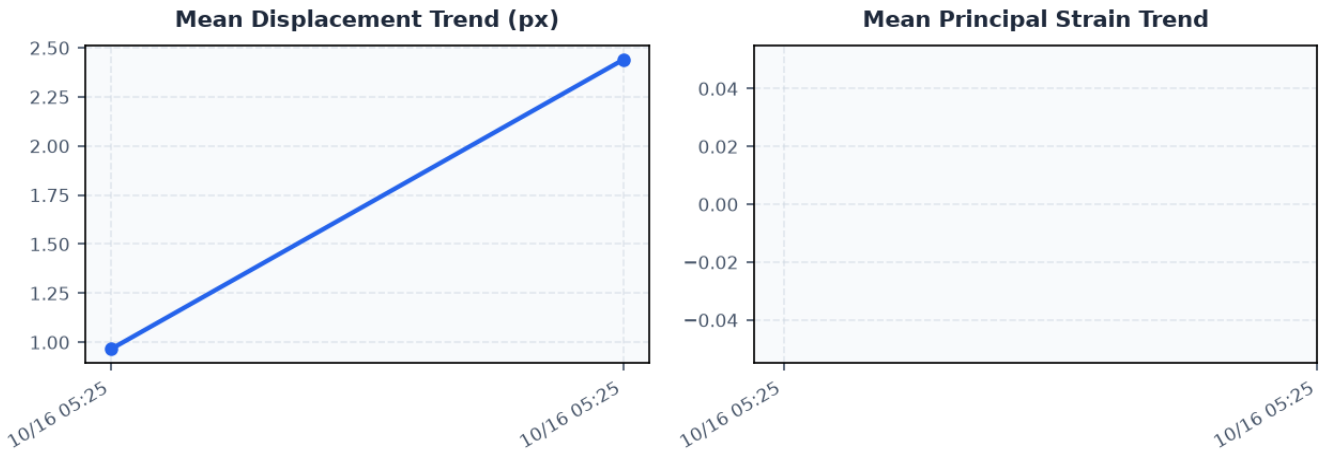
Longitudinal Multi-Session Study Analysis & Kinematic Progression Summary

Study Title:	**Multi-Session Progression Study**	**Subject Identifier:**	SUBJ-002
Study ID:	2c6c0762-a136-4ed8-8d54-17ba93756a10	**Session Count:**	**2** analyses
Baseline Date:	Oct 16, 2026 05:25	**Latest Date:**	Oct 16, 2026 05:25
Protocol Notes:	None documented.	**Generated At:**	2026-10-16 05:25 UTC

1. Longitudinal Analysis Session Timeline

Session / Date	Analysis ID	Type	Mean Disp (px)	Mean Strain	Principal Strain	QC Score
#1 • Oct 16, 2026 05:25	a0453fb6	Single Pair	0.9659	N/A	N/A	**100/100**
#2 • Oct 16, 2026 05:25	ae54ad80	Single Pair	2.4397	N/A	N/A	**100/100**

Longitudinal Kinematic Trajectories:



2. First vs. Latest Session Quantitative Delta

Deformation Metric	Baseline Value	Latest Value	Absolute Delta	Relative Change (%)	Trend Direction
Mean Displacement	0.9659 px	2.4397 px	1.4738 px	+152.6%	**Increasing**
Maximum Displacement	3.9999 px	11.0928 px	7.0929 px	+177.3%	**Increasing**
Mean Strain	N/A	N/A	N/A	—	**Insufficient_data**
Maximum Strain	N/A	N/A	N/A	—	**Insufficient_data**
Mean Principal Strain	N/A	N/A	N/A	—	**Insufficient_data**
Maximum Principal Strain	N/A	N/A	N/A	—	**Insufficient_data**
Anisotropy Index	0.3766	0.4315	0.0549	+14.6%	**Increasing**

Strain Entropy	0.8835	1.3638	0.4802	+54.4%	**Increasing**
Strain Gradient	0.0423	0.1185	0.0762	+180.2%	**Increasing**
Motion Coherence	0.9119	0.8955	-0.0163	-1.8%	**Stable**
Temporal Stability Index	N/A	N/A	N/A	—	**Insufficient_data**
ROI Mean Displacement	N/A	N/A	N/A	—	**Insufficient_data**
ROI Mean Strain	N/A	N/A	N/A	—	**Insufficient_data**
ROI Principal Strain	N/A	N/A	N/A	—	**Insufficient_data**
Computational Quality Score	100.0000 /100	100.0000 /100	0.0000 /100	+0.0%	**Stable**

3. Longitudinal Descriptive Statistics & Trajectory Trends

Tracked Metric	N (Obs)	Min	Max	Mean	Std Dev	Direction
Mean Displacement	2	0.9659	2.4397	1.7028	1.0422	**Increasing**
Maximum Displacement	2	3.9999	11.0928	7.5464	5.0154	**Increasing**
Anisotropy Index	2	0.3766	0.4315	0.4041	0.0388	**Increasing**
Strain Entropy	2	0.8835	1.3638	1.1237	0.3396	**Increasing**
Strain Gradient	2	0.0423	0.1185	0.0804	0.0539	**Increasing**
Motion Coherence	2	0.8955	0.9119	0.9037	0.0116	**Stable**
Computational Quality Score	2	100.0000	100.0000	100.0000	0.0000	**Stable**

4. Longitudinal Quality & Protocol Consistency

Mean QC Score:	**100.0 / 100**	**Quality Tiers:**	High: 2 | Mod: 0 | Low: 0
Calibration:	**UNCALIBRATED**: All sessions are uncalibrated (pixel-space).	**ROI Scope:**	**NONE**: All sessions evaluated full-image deformation fields.

Longitudinal Quality & Compatibility Warnings:
• 1 analysis ID(s) linked to this study were not found in history (they may have been cleared or deleted from the prototype cache).

Informational Notes:
• Longitudinal displacement comparisons are reported in image-space pixels because physical calibration is unavailable.

5. Technical Interpretation & Research Disclaimer

Methodological Scope: Longitudinal study analysis evaluates numerical trajectory variations in image-derived displacement fields, Green-Lagrange strain tensors, and kinematic coherence across multiple discrete analysis sessions. All trends represent algorithmic differences computed under classical continuum mechanics and dense Farnebäck optical flow.

RESEARCH ONLY — NO CLINICAL DIAGNOSTIC OR PROGRESSION ASSERTIONS:
Longitudinal analysis describes computational changes in image-derived measurements across analysis sessions. It is not a clinical diagnosis or clinical progression assessment. Categorizations such as 'increasing', 'decreasing', or 'stable' denote mathematical differentials and must NOT be interpreted as cardiac recovery, clinical deterioration, or medical efficacy. This software is an experimental research prototype and must NOT be used as a medical diagnostic device.